

P. Shubham Parashar

THEORETICAL PHYSICIST

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RESEARCH FOCUS

Theoretical condensed-matter physicist working on quantum transport, kinetic theory, electron hydrodynamics, magneto-transport, and quantum Hall systems. I study a general inference problem: when finite-resolution maps of voltage, current, and dissipation can distinguish microscopic kinetic dynamics from an effective fluid description.

SELECTED PUBLICATIONS AND CURRENT MANUSCRIPTS

CURRENT MANUSCRIPTS

Complete kinematic null for local kinetic dissipation in a sixfold-driven electron fluid. P. Shubham Parashar.

Under review at *Physica B* (2026). [arXiv:2608.10031](#).

Fourth-order closure obstruction and chiral nonlocality in circular kinetic magnetotransport. P. Shubham Parashar.

Under review at *Physical Review B* (2026). [arXiv:2607.22730](#).

Thermal and viscous contrast in quantum Hall scanning-probe images. P. Shubham Parashar.

Under review at *Physica E* (2026). [arXiv:2607.16315](#).

Diagnosing changes in trajectory-geometry tails: endpoint laws, crossovers, and finite-window tests. P. Shubham Parashar.

Under review at *The European Physical Journal B* (2026).

REFEREED PUBLICATIONS

Diffusive and hydrodynamic magnetotransport around a density perturbation in a two-dimensional electron gas. P. Shubham Parashar and M. M. Fogler.

Physical Review B **113**, 155443 (2026). [arXiv:2601.16402](#).

Nanoscale imaging of magnetotransport around a circular p–n junction in graphene. Z. J. Krebs *et al.*, including P. Shubham Parashar.

Physical Review Letters **136**, 136301 (2026), Editors' Suggestion. [arXiv:2409.19468](#).

EDUCATION AND RESEARCH

2022–2026 **Ph.D., Physics**, University of California San Diego. Advisor: Michael M. Fogler.

2017–2022 **Researcher**, University of California San Diego. Research with Benjamin Grinstein and Congjun Wu.

2016–2017 **M.S., Physics**, Indian Institute of Science, Bangalore. Advisor: Rahul Pandit.

2012–2016 **B.S., Physics**, Indian Institute of Science, Bangalore.

SELECTED RESEARCH CAPABILITIES

- **Analytical kinetic transport theory.** Angular-harmonic expansions of the Boltzmann equation, higher-gradient and finite-wave-number response, and controlled hydrodynamic closures.
- **Symmetry-designed observables.** Drive and device symmetries that suppress leading dissipative channels locally and isolate higher kinetic harmonics as measurable signals.
- **Finite-resolution transport inference.** Spatial voltage, current, and dissipation maps, thermoelectric and Hall contrast, and finite-resolution scanning-probe inference.

METHODS

Boltzmann kinetic theory · electron hydrodynamics · semiclassical and quantum transport · linear response · asymptotics · spatial transport inference · numerical modeling · Python · Mathematica · LaTeX

DISTINCTIONS

Bhadra Fellowship (2024) · UCSD Scholar's Grant (2017) · KVPY Fellowship (2012–2017)